SHEET INDEX

SHEET NUMBER	SHEET NAME
GENERAL	
GI000	COVER SHEET
ELECTRICAL	
E001	INDEX, LEGEND, AND GENERAL NOTES
E103	BUILDING 103 PLAN
E104	BUILDING 104 PLAN
E203	BUILDING 103 SINGLE LINE
E204	BUILDING 104 SINGLE LINE
E301	ROOF PATHING PLANS

VA ILLIANA HEALTH CARE SYSTEM (VAIHCS)

1900 E. MAIN ST., DANVILLE, IL

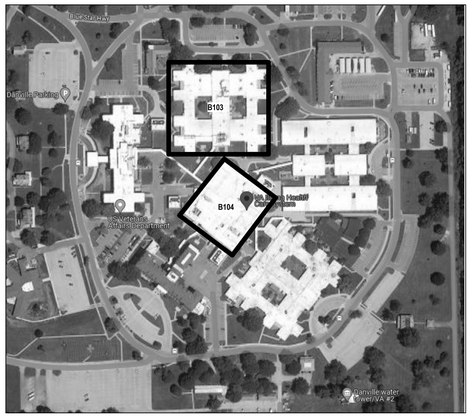
BID DOCUMENTS

INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS

550-23-104

AREA MAP

VICINITY MAP



CAMPUS MAP

OWNER

VA Illiana Health Care System

Contact:
Michael Wlos
COR

1900 E. Main St.
Danville, IL 61832

p: 217.554.5985
e: michael.wlos@va.gov

PROJECT MANAGER

GUIDON DESIGN INC.

Contact:
Tory McNeil
Project Manager

1221 North Pennsylvania Street
Indianapolis, IN 46202

p: 317.800.6388
e: tmcneil@guidondesign.com

ELECTRICAL

Heapy Engineering

Contact:
Matt Graham
Electrical Engineer

3850 Priority Way South Dr
Indianapolis, IN 46240

p: 937-224-0861
e: msgraham@heapy.com

Revisions:	Date:

CONSULTANT

Consultant: HEAPY

P: 937-224-0861 www.heapy.com
PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD

A/E:
GUIDON
1221 N. PENNSYLVANIA STREET
INDIANAPOLIS, IN 46202
317.800.6388

STAMP

19-5002.17
TORY MCNEIL
REGISTERED
No. PE11900522
STATE OF INDIANA
PROFESSIONAL ENGINEER
11/15/2023

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
COVER SHEET	BID DOCUMENTS	INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	550-23-104
			Building Number: B103 & B104
Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Drawing Number: GI000
		Issue Date: 11/15/2023 · Checked: · Drawn:	

VA FORM 08 - 6231

Autodesk Docs://19-5002.17 - Install Rooftop Photovoltaic/19.5002.17_A_v22.rvt

11/15/2023 11:03:54 AM

GENERAL NOTES

A. THESE NOTES APPLY EQUALLY TO THE FULL SET OF DOCUMENTS.

B. ALL ABANDONED EXTRANEOUS CONDUITS, WIRING, DEVICES, ETC. SHALL BE REMOVED.

C. INCLUDE ALL WORK NECESSARY TO ACCOMMODATE PHASING. REFER TO ARCHITECTURAL DRAWINGS AND GENERAL REQUIREMENTS SECTION 01 00 00.

D. THE CONTRACTOR SHALL REFER TO ALL SPECIFICATION SECTIONS AND THE DRAWINGS FOR DETAILS OF BUILDING CONSTRUCTION TO ENSURE SPACE AND SATISFACTORY ARRANGEMENT FOR THEIR WORK. THE VARIOUS DRAWINGS COMPRISING THE SET ARE INTERDEPENDENT AND MUST BE USED JOINTLY AT ALL TIMES. THE CONTRACTOR SHALL REFER TO THE GENERAL REQUIREMENTS OF THE CONTRACT. THE NOTES AND SYMBOLS INDICATED ON THE DRAWINGS ARE FOR THE GUIDANCE OF ALL TRADES INVOLVED IN THE PROJECT AND MUST BE FOLLOWED TO EXECUTE THE WORK AS INTENDED. IF DISCREPANCIES OCCUR, CONTACT THE COR THROUGH THE CONTRACTING OFFICER FOR CLARIFICATION BEFORE PROCEEDING.

E. ALL WORK MUST BE COORDINATED WITH THE CONTRACTING OFFICER AND (FOR ALL OTHERS) TO MAINTAIN OPERATION OF THE EXISTING FACILITY.

F. ALL WORK SHALL BE PERFORMED IN COMPLETE COMPLIANCE WITH ALL CURRENT GOVERNING CODES AND STANDARDS.

G. EXISTING CONDITIONS SHOWN HAVE BEEN BASED UPON AVAILABLE DRAWING INFORMATION, AND MAY BE AT VARIANCE WITH ACTUAL WORK IN PLACE. THE CONTRACTOR SHALL TAKE ALL NECESSARY FIELD MEASUREMENTS AND FIELD VERIFY ALL CONDITIONS AFFECTING THE EXECUTION OF THE WORK. ANY DISCREPANCIES BETWEEN EXISTING CONDITIONS AND THE WORK SHOWN ON THE CONTACT DOCUMENTS WHICH MAY IMPACT THE PROGRESS OF THE WORK SHALL BE BROUGHT TO THE ATTENTION OF THE CONTRACTING OFFICER IN WRITING FOR RESOLUTION BEFORE PROCEEDING WITH THE WORK.

H. ALL PENETRATIONS THROUGH FIRE-RATED ASSEMBLIES SHALL BE PROJECTED AND/OR FIRE-STOPPED AS REQUIRED TO MAINTAIN FIRE-RATINGS INDICATED. COORDINATE WITH ALL TRADES TO ENSURE FIRE-RATED PENETRATION REQUIREMENTS AND DETAILS ARE MET.

I. ANNULAR SPACE OF ALL PIPE, CONDUIT, DUCT AND OTHER SIMILAR PENETRATIONS OF FIRE RATED ASSEMBLIES SHALL BE FIRESTOPPED. IN ADDITION, PENETRATIONS THROUGH 0-HOUR RATED WALLS AND FLOORS SHALL BE FIRESTOPPED TO RETARD PASSAGE OF FIRE AND SMOKE.

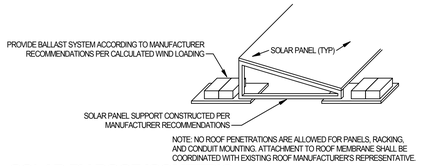
J. ALL CONDUITS IN FINISHED ROOMS AND CORRIDORS SHALL BE CONCEALED IN A WALL OR ABOVE CEILING.

K. ACCESS PANELS IN NON ACCESSIBLE CEILINGS SHALL BE PROVIDED FOR ALL ELECTRICAL ITEMS REQUIRING ACCESS.

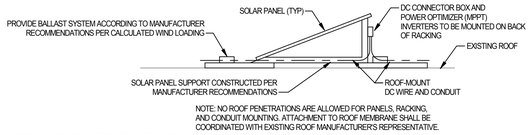
L. ALL CUTTING AND PATCHING REQUIRED FOR THIS PROJECT SHALL BE INCLUDED IN THE CONTRACT. REFINISH ANY SURFACE DISTURBED UNDER THIS WORK TO MATCH EXISTING.

M. ANY REMOVED EQUIPMENT SHALL BE TURNED OVER TO THE VA. ITEMS NOT DESIRED BY THE VA SHALL BE REMOVED FROM THE PREMISES AND DISPOSED OF PROPERLY BY THE CONTRACTOR.

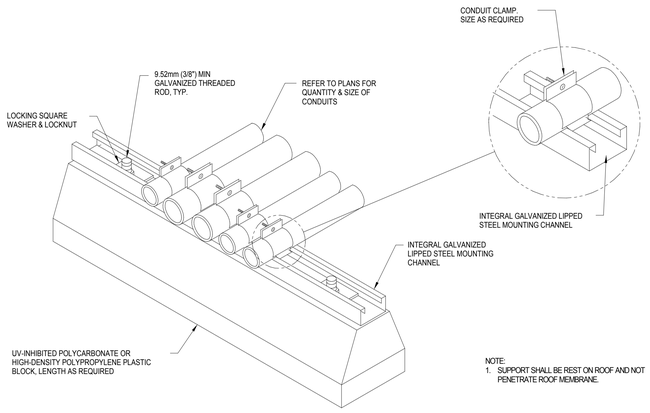
N. THE CONTRACT DRAWINGS ARE NOT INTENDED TO SHOW EVERY VERTICAL OR HORIZONTAL OFFSET WHICH MAY BE NECESSARY TO COMPLETE THE SYSTEMS. COORDINATE WORK IN ADVANCE WITH ALL OTHER TRADES AND REPORT IMMEDIATELY ANY DIFFICULTIES WHICH CAN BE ANTICIPATED.



1 SOLAR PV ROOF ISOMETRIC
SCALE: NONE

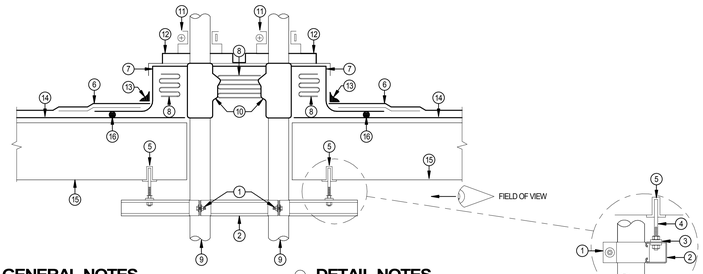


2 SOLAR PV ROOF RACK
SCALE: NONE



NOTE:
1. SUPPORT SHALL BE REST ON ROOF AND NOT PENETRATE ROOF MEMBRANE.

3 ROOF CONDUIT SUPPORT DETAIL
SCALE: NONE



GENERAL NOTES

1. MAINTAIN A MINIMUM CLEARANCE OF 1'-0" [305mm] ON ALL SIDES OF ROOF PENETRATION FROM WALLS, CURBS AND OTHER PROJECTION TO FACILITATE PROPER FLASHING.
2. FLANGES OF ADJACENT FLASHING SHALL NOT BE CUT OR OVERLAPPED.
3. VERIFY ROOF AND STRUCTURAL SYSTEM WITH ARCHITECT.
4. COORDINATE FLASHING INSTALLATION WITH ROOFING MANUFACTURER TO ENSURE PROPER METHODS AND MATERIALS ARE USED TO MAINTAIN ROOF WARRANTY.
5. VERIFY EXISTING ROOF CONSTRUCTION MATCHES DETAIL. BRING ANY VARIANCES TO THE ATTENTION OF THE ARCHITECT/ENGINEER PRIOR TO COMMENCING CONSTRUCTION.

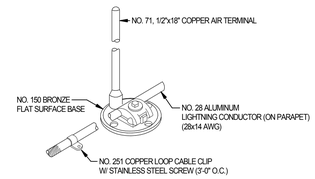
DETAIL NOTES

1. CONDUIT CLAMP. SIZE AS REQUIRED.
2. INTEGRAL GALVANIZED LIPPED STEEL MOUNTING CHANNEL.
3. LOCKING SQUARE & LOCKNUT.
4. 3/8" [9.52mm] MIN. GALVANIZED THREADED ROD.
5. STEEL EXPANSION ANCHOR.
6. MULTIPLE MEMBRANE STRIPPING PLIES FEATHERED AT FIELD OF ROOF BY ROOFING CONTRACTOR. PRIME FLANGE BEFORE STRIPPING.
7. PREFABRICATED FORMED 0.06" [1.52mm] THICK ALUMINUM FLASHING ROUND COLLAR WITH MINIMUM 3" [76mm] WIDE FLANGE. ATTACHED TO ROOF PER MANUFACTURER'S INSTRUCTIONS.
8. BATT INSULATION.
9. CONDUIT.
10. SEAL -OFF FITTING.
11. STAINLESS STEEL CLAMP.
12. PREFABRICATED RUBBER BOOT CAP WITH GRADUATED STEPS & WEATHER-ROOF PRESSURE SEAL TO COLLAR.
13. SEALANT AT PENETRATION TO MEMBRANE JUNCTURE BY ROOFING CONTRACTOR.
14. ROOF MEMBRANE.
15. CONCRETE DECKING.
16. WATER CUT-OFF MASTIC COMPATIBLE WITH ROOF MEMBRANE.

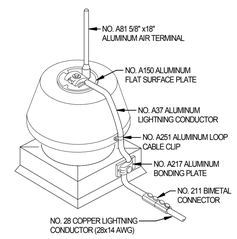
4 CONDUIT ROOF PENETRATIONS
SCALE: NONE

ELECTRICAL SYMBOLS

3	DASH SYMBOL INDICATES PARTICULAR OUTLET OR DEVICE TO BE REMOVED AND CIRCUITRY MADE CONTINUOUS WHERE REQUIRED.
3	EXISTING OUTLET OR DEVICE TO REMAIN. MAINTAIN EXISTING CIRCUITING.
	ELECTRICAL CONNECTION.
	20A-125V DUPLEX RECEPTACLE, NEMA 5-20R (18" MH UNLESS NOTED OTHERWISE). WHEN ⏻ SHOWN, RECEPTACLE TO HAVE "CONTROLLED" MARKINGS.
GF	20A-125V DUPLEX RECEPTACLE, NEMA 5-20R, WITH GROUND FAULT CIRCUIT INTERRUPTER (18" MH UNLESS NOTED OTHERWISE).
WP/GF	20A-125V WEATHERPROOF DUPLEX RECEPTACLE, NEMA 5-20R WITH GROUND FAULT CIRCUIT INTERRUPTER (18" MH UNLESS NOTED OTHERWISE), WITH IN-USE WEATHERPROOF EXTRA DUTY GRAY COVER, VERTICAL MOUNT.
	ELECTRICAL PANEL OR SWITCHBOARD PER DRAWINGS.
J	DISCONNECT SWITCH.
	ROOF RACK-MOUNT PV SOLAR PANEL.



5 LIGHTNING PROTECTION DETAILS
SCALE: NONE



6 LIGHTNING PROTECTION DETAILS
SCALE: NONE

ABBREVIATIONS

AAP	- AREA ALARM PANEL - MEDICAL GAS
ACC	- ACCESS
ADJ	- ADJUSTABLE
AF	- ARC FAULT CIRCUIT INTERRUPTER
AFCI	- ARC FAULT CIRCUIT INTERRUPTER
AFF	- ABOVE FINISHED FLOOR TO BOTTOM OF ITEM
AFG	- ABOVE FINISHED GRADE TO BOTTOM OF ITEM
ALT	- ALTERNATE
AP	- ACCESS PANEL
APPROX	- APPROXIMATE
ARCH	- ARCHITECT OR ARCHITECTURAL
ASSY	- ASSEMBLY
ATS	- AUTOMATIC TRANSFER SWITCH
BLDG	- BUILDING
BOE	- BOTTOM OF EQUIPMENT
BOT	- BOTTOM
BTWN	- BETWEEN
CFCI	- CONTRACTOR FURNISHED CONTRACTOR INSTALLED
CKT	- CIRCUIT
CLG	- CEILING
CMU	- CONCRETE MASONRY UNIT
CONN	- CONNECT OR CONNECTION
CONTR	- CONTRACTOR
COR	- CONTRACTING OFFICER'S REPRESENTATIVE
CORR	- CORRIDOR
CTR	- CENTER
D	- DEPTH
DET	- DETAIL
DIA	- DIAMETER
DIM	- DIMENSION
DIV	- DIVISION
DN	- DOWN
DWG	- DRAWING
EA	- EACH
EC	- ELECTRICAL (DIVISION 26)
EJ	- EXPANSION JOINT
ELEC	- ELECTRICAL
ELEV	- ELEVATION OR ELEVATOR
EM	- EMERGENCY
EQ	- EQUAL
EQS	- EQUIPMENT SUPPLIER
EQUIP	- EQUIPMENT
ETR	- EXISTING TO REMAIN
EX	- EXISTING
EXP	- EXPANSION
EXT	- EXTERIOR
FCE	- FIRE CONTROL EQUIPMENT
FF	- FINISHED FLOOR ELEVATION
FLR	- FLOOR
FSC	- FIRE SUPPRESSION (DIVISION 21)
FT	- FEET
FTG	- FOOTING
GC	- GENERAL TRADES (DIVISION 01)
GEC	- GROUNDING ELECTRODE CONDUCTOR
GF	- GROUND FAULT CIRCUIT INTERRUPTER
GFCI	- GROUND FAULT CIRCUIT INTERRUPTER OR GOVERNMENT FURNISHED CONTRACTOR INSTALLED
GFFT	- GROUND FAULT FEED THRU
GFP	- GROUND FAULT PROTECTION
HC	- HVAC (DIVISION 23)
HP	- HORSE POWER OR HIGH POINT
HVAC	- HEATING, VENTILATING, AND AIR CONDITIONING
ID	- INSIDE DIAMETER
IN	- INCHES
KEC	- KITCHEN EQUIPMENT CONTRACTOR
L	- LENGTH
LBS	- POUNDS
MAP	- MASTER ALARM PANEL (MEDICAL GAS)
MAX	- MAXIMUM
MEZZ	- MEZZANINE
MFR	- MANUFACTURER
MH	- MANHOLE OR MOUNTING HEIGHT TO CENTER LINE OF ITEM
MIN	- MINIMUM OR MINUTE
MISC	- MISCELLANEOUS
MPPT	- MAXIMUM POWER POINT TRACKING (POWER OPTIMIZER)
MTD	- MOUNTED
MTG	- MOUNTING
NIC	- NOT IN CONTRACT
NOM	- NOMINAL
NTS	- NOT TO SCALE
OD	- OUTSIDE DIAMETER
OFCI	- OWNER FURNISHED CONTRACTOR INSTALLED
OFOI	- OWNER FURNISHED OWNER INSTALLED
PC	- PLUMBING (DIVISION 22)
PLBG	- PLUMBING
RAD	- RADIUS
REC	- RECESSED
REQD	- REQUIRED
RI	- ROUGH-IN
S	- SURFACE MOUNTED
SC	- SECURITY (DIVISION 28)
SCH	- SCHEDULE
SHT	- SHEET
SMS	- SECURITY MANAGEMENT SYSTEM
SPEC	- SPECIFICATIONS
SQ	- SQUARE
SS	- STAINLESS STEEL
STD	- STANDARD
STRUC	- STRUCTURAL OR STRUCTURE
SUC	- SITE UTILITY CONTRACTOR
TC	- TECHNOLOGY (DIVISION 27)
TEMP	- TEMPERATURE
TOE	- TOP OF EQUIPMENT
TYP	- TYPICAL
UNO	- UNLESS NOTED OTHERWISE
VFD	- VARIABLE FREQUENCY DRIVE
VOL	- VOLUME
W/	- WITH
W/O	- WITHOUT
WP	- WEATHERPROOF
ZVC	- ZONE VALVE CABINET

GENERAL FLOOR PLAN NOTES

B / E2	DETAIL: B = DETAIL DESIGNATION E2 = SHEET WHERE DETAIL IS LOCATED
1 / E2	SECTION: 1 = SECTION DESIGNATION E2 = SHEET WHERE SECTION IS LOCATED
T2 1	ELEVATION: 1 = ELEVATION DESIGNATION T2 = SHEET WHERE ELEVATION IS LOCATED
3	PLAN NOTE. APPLIES ONLY TO THE SHEET WHICH IT IS SHOWN.
3	DETAIL NOTE. APPLIES ONLY TO THE ASSOCIATED DETAIL.
	WIRE & CONDUIT IN WALL OR ABOVE CEILING.
	WIRE & CONDUIT IN OR BELOW SLAB OR GRADE.
	CONDUIT TO BE REMOVED.
EX	EXISTING WIRE & CONDUIT TO REMAIN.
X - 1,2	EACH ARROWHEAD REPRESENTS ONE COMPLETE CIRCUIT; "X" DENOTES PANEL NAME; NUMBER(S) DENOTES CIRCUIT(S).

NOTE: ALL SYMBOLS AND ABBREVIATIONS ARE SUBJECT TO MODIFICATIONS ON OTHER DRAWINGS.

ALL SYMBOLS OR ABBREVIATIONS MIGHT NOT NECESSARILY BE USED ON THIS PROJECT.

ELECTRICAL SHEET LIST

SHEET NUMBER	SHEET NAME
E001	INDEX, LEGEND AND GENERAL NOTES
E103	BUILDING 103 PLAN
E104	BUILDING 104 PLAN
E203	BUILDING 103 SINGLE LINE
E204	BUILDING 104 SINGLE LINE
E301	ROOF PATHING PLANS

Revisions: Date:

CONSULTANT

Consultant: HEAPY
P: 937-224-0861 www.heapy.com
PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD

A/E:
GUIDON
1221 N. PENNSYLVANIA STREET
INDIANAPOLIS, IN 46202
317.800.6388

STAMP

LICENSED PROFESSIONAL ENGINEER MATTHEW S. GRAHAM 062-073523 STATE OF ILLINOIS 11/15/2023

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title: INDEX, LEGEND AND GENERAL NOTES	Phase: BID DOCUMENTS	Project Title: INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	Project Number: 550-23-104
			Building Number: B103 & B104
Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Drawing Number: E001
		Issue Date: 10/06/2023 · Checked: JAS · Drawn: NES	

VA FORM 08 - 6231

Autodesk Docs://19.5092.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt 11/15/2023 5:07:27 PM

GENERAL NOTES:

A. OPEN BOXED AREAS SHOWN INDICATE ROOF MOUNTED OBSTRUCTIONS. THE DASHED OUTLINE REPRESENTS A 4' OFFSET FROM THE EDGE OF THE OBSTRUCTION. CONTRACTOR SHALL FEILD MEASURE AND COORDINATE PANEL LAYOUT WITH THE ROOF MOUNTED EQUIPMENT. PROVIDE OPEN WALKWAYS TO THIS EQUIPMENT TO FACILATE MAINTENANCE OF THIS EQUIPMENT. ALSO PROVIDE WALKING SPACE BETWEEN ROWS FOR PANEL CLEANING OR MAINTENANCE.

B. SOLAR PANEL SYSTEM LAYOUT SHOWN IS ONLY A REPRESENTATION OF WHERE PANELS SHALL BE PLACED. CONTRACTOR SHALL PROVIDE FINAL SOLAR PANEL ARRAY LAYOUT ALONG WITH SIGNED AND SEALED STRUCTURAL WIND LOADING CALCULATIONS FOR THE BALLASTED DESIGN.

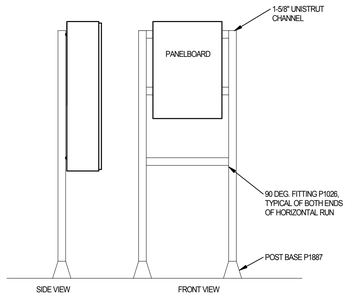
C. CONTRACTOR SHALL LOCATE THE SOLAR PANEL SYSTEM INVERTERS TO BE READILY ACCESSIBLE TO A WALKWAY.

D. MAINTAIN EXISTING LIGHTING PROTECTION ON ROOF AND ROOF PARAPET. CONNECT SOLAR PANEL RACKING EXISTING LIGHTNING PROTECTION CROSS MEMBER CABLING PER MANUFACTURER REQUIREMENTS. COORDINATE LOCATION AND NUMBER OF NEW/REPLACEMENT AIR TERMINALS FOR A COMPLETE UL 1703 LIGHTING PROTECTION SYSTEM.

E. REFERENCE SHEET E301 FOR WALKING PATH INFORMATION.

NOTES:

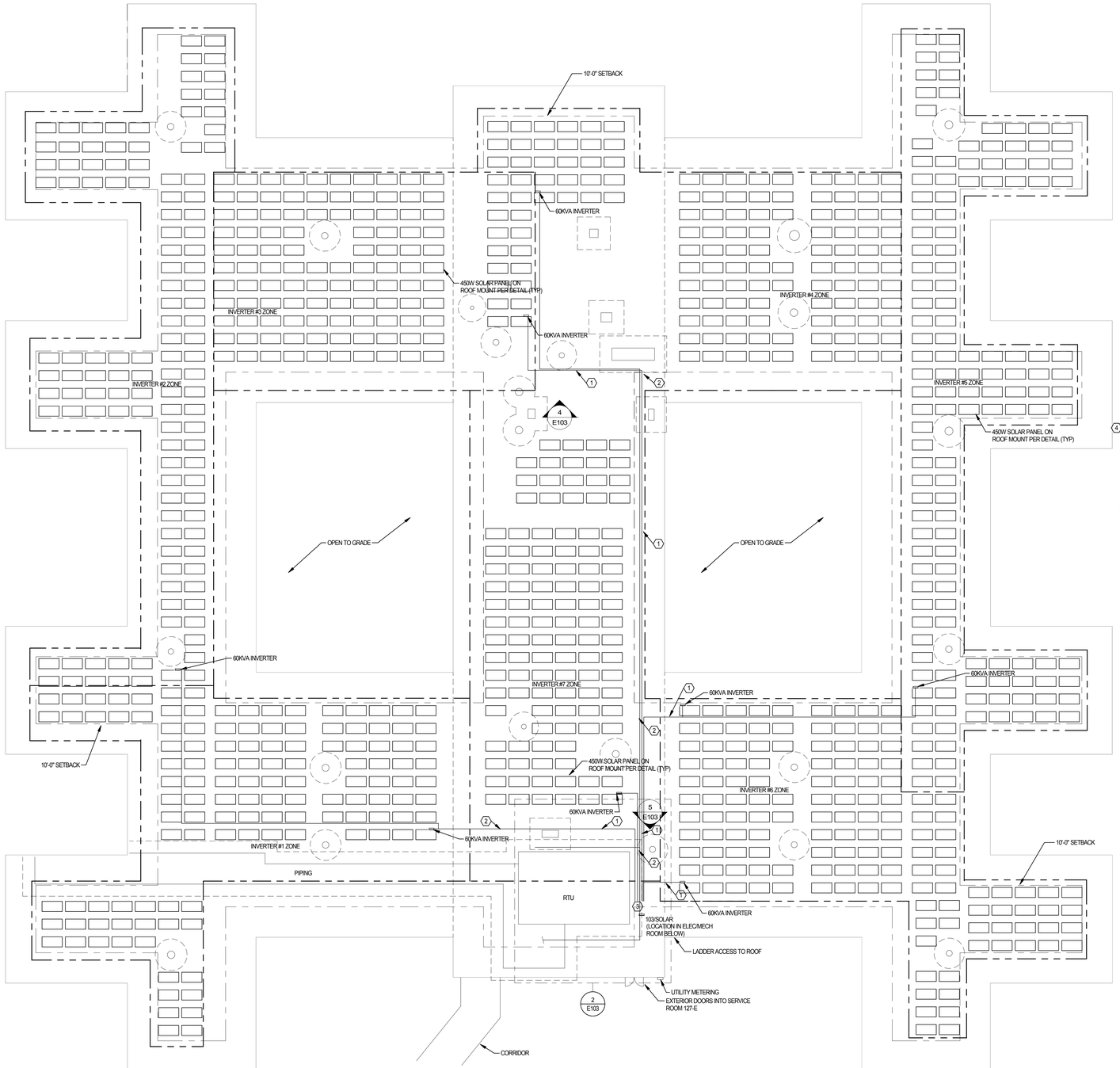
1. AC FEEDER, ROUTE FEEDER BETWEEN INVERTER AND DISTRIBUTION PANEL 103/SOLAR. PROVIDE ROOF BLOCKING SUPPORTS ALONG ROUTING EVERY 6'.
2. PROVIDE CROSS OVER BRIDGING WHERE AC FEEDERS CROSS WALKING PATH.
3. AC FEEDER DOWN TO PANEL 103/SOLAR. COORDINATE CONDUIT ROOF PENETRATION WITH ROOFING CONTRACTOR FOR WARRANTY COMPLIANCE. REFER TO ROOF PENETRATION DETAIL ON SHEET E001.
4. PROVIDE NEW BREAKER TO SERVE PANEL 103/SOLAR. REFER TO SINGLE LINE DIAGRAM.
5. PROVIDE METERING TO BE MONITORED BY BAS AND CLIENT eDART SYSTEM. PROVIDE ALL REQUIRED ACCESSORIES AND PROGRAMMING. ALL WORK REQUIRING SHUT-DOWN OF ELECTRICAL SERVICES ARE TO BE PERFORMED ON A WEEKEND DAY COORDINATED WITH COR.
6. UTILITY METERING. COORDINATE ROUGH-IN REQUIREMENTS WITH UTILITY PROVIDER.



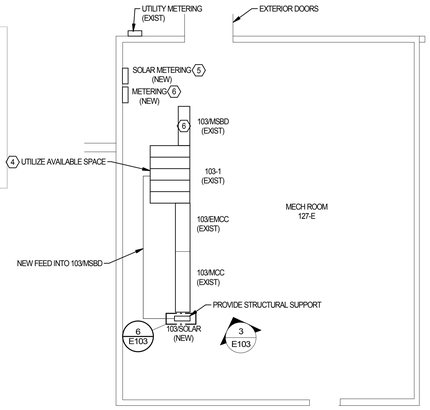
6 **FREE STANDING PANEL**
SCALE: NONE

4 **BUILDING 103 NORTH VIEW**
SCALE: NONE

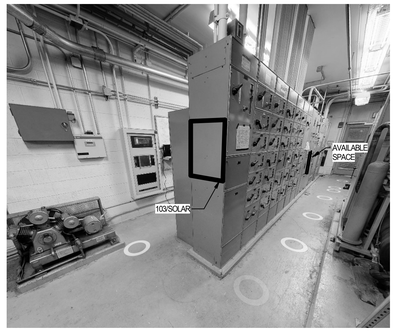
5 **BUILDING 103 SOUTH VIEW**
SCALE: NONE



1 **BUILDING 103 ROOF PLAN**
SCALE: 1/16" = 1'-0"



2 **FIRST FLOOR - SERVICE RM 127-E**
SCALE: 1/8" = 1'-0"



3 **BUILDING 103 MAIN ELEC RM 127-E**
SCALE: NONE

NOTE: LAYOUT SHOWN FOR PRICING AND GENERAL INFORMATION ONLY. CONTRACTOR SHALL PROVIDE LAYOUT PLAN BASED ON APPROVED MANUFACTURER'S EXACT PANEL DIMENSIONS AND DC STRING RECOMMENDATIONS. PLAN SHALL SHOW PANEL LOCATIONS, DC CONDUIT PATHING, AND MPPT MOUNTING LOCATIONS. LAYOUT SHALL ALLOW MAINTENANCE ACCESS AND CLEARANCE TO ALL ROOF EQUIPMENT. EXISTING LIGHTNING PROTECTION SYSTEM SHALL BE MODIFIED AS REQUIRED TO INSTALL THE SOLAR POWER SYSTEM.

Revisions:	Date:

CONSULTANT

Consultant: HEAPY
P: 937-224-0861 www.heapy.com
PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD

A/E:
GUIDON
1221 N. PENNSYLVANIA STREET
INDIANAPOLIS, IN 46202
317.800.6388

STAMP

LICENSED PROFESSIONAL ENGINEER
MATTHEW S. GRAHAM
062-073523
STATE OF ILLINOIS
10/06/2023

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BUILDING 103 PLAN	BID DOCUMENTS	INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	550-23-104
			Building Number: B103 & B104
Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Drawing Number: E103
		Issue Date: 10/06/2023 · Checked: JAS · Drawn: NES	

VA FORM 08 - 6231

Autodesk Docs://19.5002.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt
11/15/2023 5:07:38 PM

GENERAL NOTES:

A. OPEN BOXED AREAS SHOWN INDICATE ROOF MOUNTED OBSTRUCTIONS. THE DASHED OUTLINE REPRESENTS A 4' OFFSET FROM THE EDGE OF THE OBSTRUCTION. CONTRACTOR SHALL FEILD MEASURE AND COORDINATE PANEL LAYOUT WITH THE ROOF MOUNTED EQUIPMENT. PROVIDE OPEN WALKWAYS TO THIS EQUIPMENT TO FACILATE MAINTENANCE OF THIS EQUIPMENT. ALSO PROVIDE WALKING SPACE BETWEEN ROWS FOR PANEL CLEANING OR MAINTENANCE.

B. SOLAR PANEL SYSTEM LAYOUT SHOWN IS ONLY A REPRESENTATION OF WHERE PANELS SHALL BE PLACED. CONTRACTOR SHALL PROVIDE FINAL SOLAR PANEL ARRAY LAYOUT ALONG WITH SIGNED AND SEALED STRUCTURAL WIND LOADING CALCULATIONS FOR THE BALLASTED DESIGN.

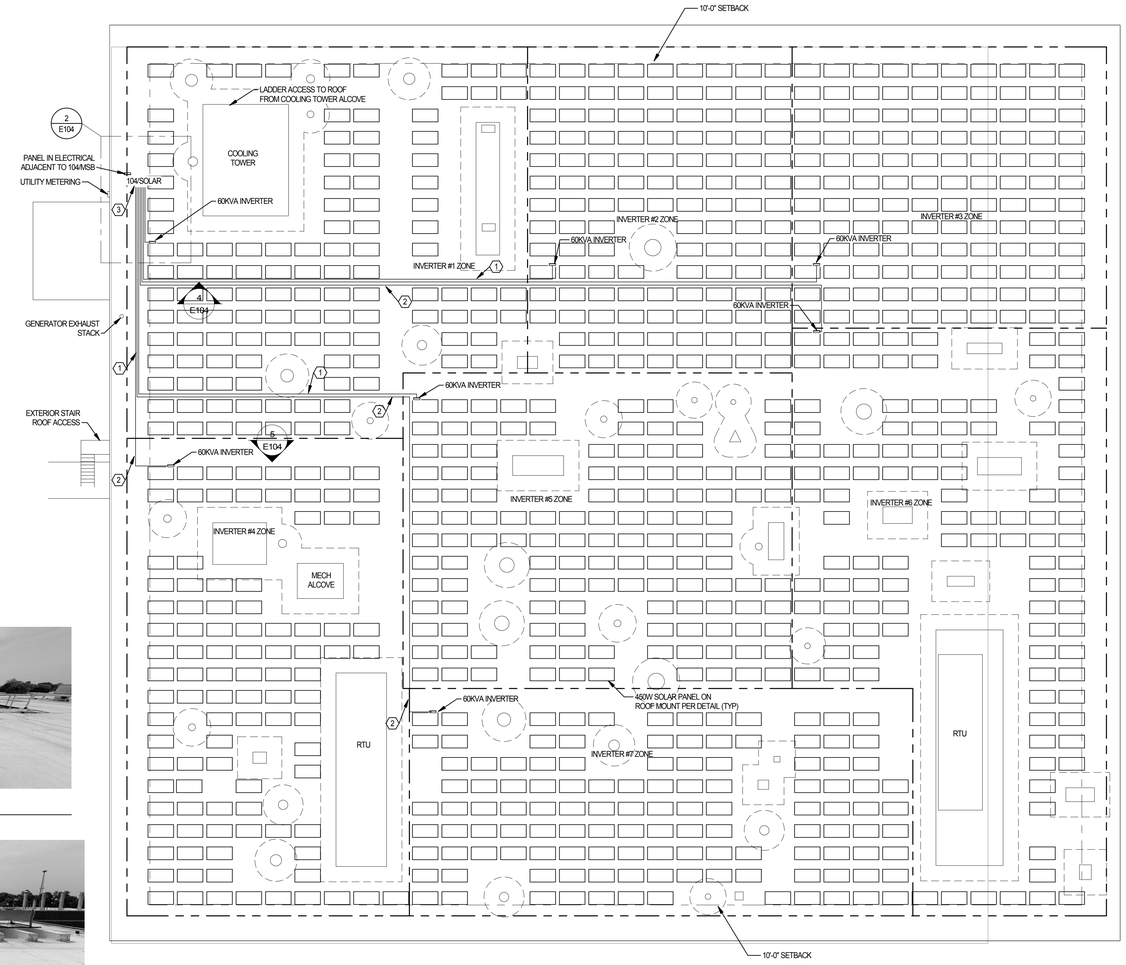
C. CONTRACTOR SHALL LOCATE THE SOLAR PANEL SYSTEM INVERTERS TO BE READILY ACCESSIBLE TO A WALKWAY.

D. MAINTAIN EXISTING LIGHTNING PROTECTION ON ROOF AND ROOF PARAPET. CONNECT SOLAR PANEL RACKING EXISTING LIGHTNING PROTECTION CROSS MEMBER CABLING PER MANUFACTURER REQUIREMENTS. COORDINATE LOCATION AND NUMBER OF NEW/REPLACEMENT AIR TERMINALS FOR A COMPLETE UL 1703 LIGHTING PROTECTION SYSTEM.

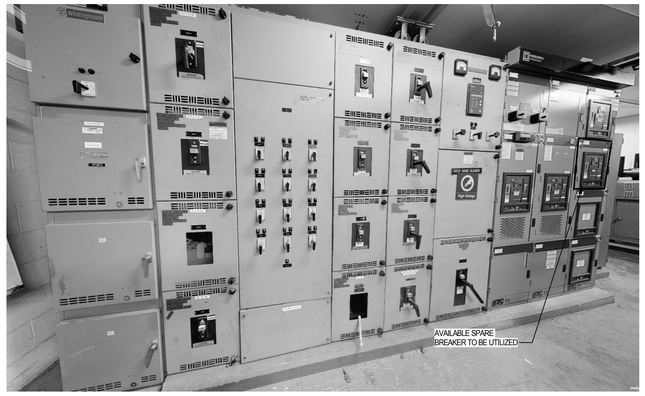
E. REFERENCE SHEET E301 FOR WALKING PATH INFORMATION.

NOTES:

1. AC FEEDER. ROUTE FEEDER BETWEEN INVERTER AND DISTRIBUTION PANEL 104/SOLAR. PROVIDE ROOF BLOCKING SUPPORTS ALONG ROUTING EVERY 6'.
2. PROVIDE CROSS OVER BRIDGING WHERE AC FEEDERS CROSS WALKING PATH.
3. AC FEEDER DOWN TO PANEL 104/SOLAR. COORDINATE CONDUIT ROOF PENETRATION WITH ROOFING CONTRACTOR FOR WARRANTY COMPLIANCE. REFER TO ROOF PENETRATION DETAIL ON SHEET E001.
4. UTILIZE SPARE BREAKER TO SERVE PANEL 104/SOLAR. REFER TO SINGLE LINE DIAGRAM.
5. PROVIDE METERING TO BE MONITORED BY BAS AND CLIENT eDART SYSTEM. PROVIDE ALL REQUIRED ACCESSORIES AND PROGRAMMING. ALL WORK REQUIRING SHUT-DOWN OF ELECTRICAL SERVICES ARE TO BE PERFORMED ON A WEEKEND DAY COORDINATED WITH COR.
6. UTILITY METERING. COORDINATE ROUGH-IN REQUIREMENTS WITH UTILITY PROVIDER.



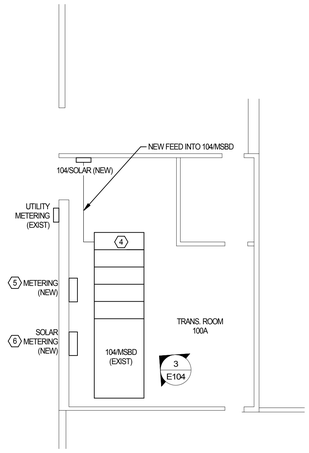
1 BUILDING 104 ROOF PLAN
SCALE: 1/16" = 1'-0"



3 BUILDING 104 MAIN ELEC ROOM RM 100A
SCALE: NONE

4 BUILDING 104 NORTH VIEW
SCALE: NONE

5 BUILDING 104 SOUTH VIEW
SCALE: NONE



2 FIRST FLOOR - TRANSFORMER RM 100A
SCALE: 1/8" = 1'-0"

NOTE: LAYOUT SHOWN FOR PRICING AND GENERAL INFORMATION ONLY. CONTRACTOR SHALL PROVIDE LAYOUT PLAN BASED ON APPROVED MANUFACTURER'S EXACT PANEL DIMENSIONS AND DC STRING RECOMMENDATIONS. PLAN SHALL SHOW PANEL LOCATIONS, DC CONDUIT PATHING, AND MPPT MOUNTING LOCATIONS. LAYOUT SHALL ALLOW MAINTENANCE ACCESS AND CLEARANCE TO ALL ROOF EQUIPMENT. EXISTING LIGHTNING PROTECTION SYSTEM SHALL BE MODIFIED AS REQUIRED TO INSTALL THE SOLAR POWER SYSTEM.

Revisions:	Date:

CONSULTANT

Consultant: HEAPY
P: 937-224-0861 www.heapy.com
PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD
A/E:
GUIDON
1221 N. PENNSYLVANIA STREET
INDIANAPOLIS, IN 46202
317.800.6388

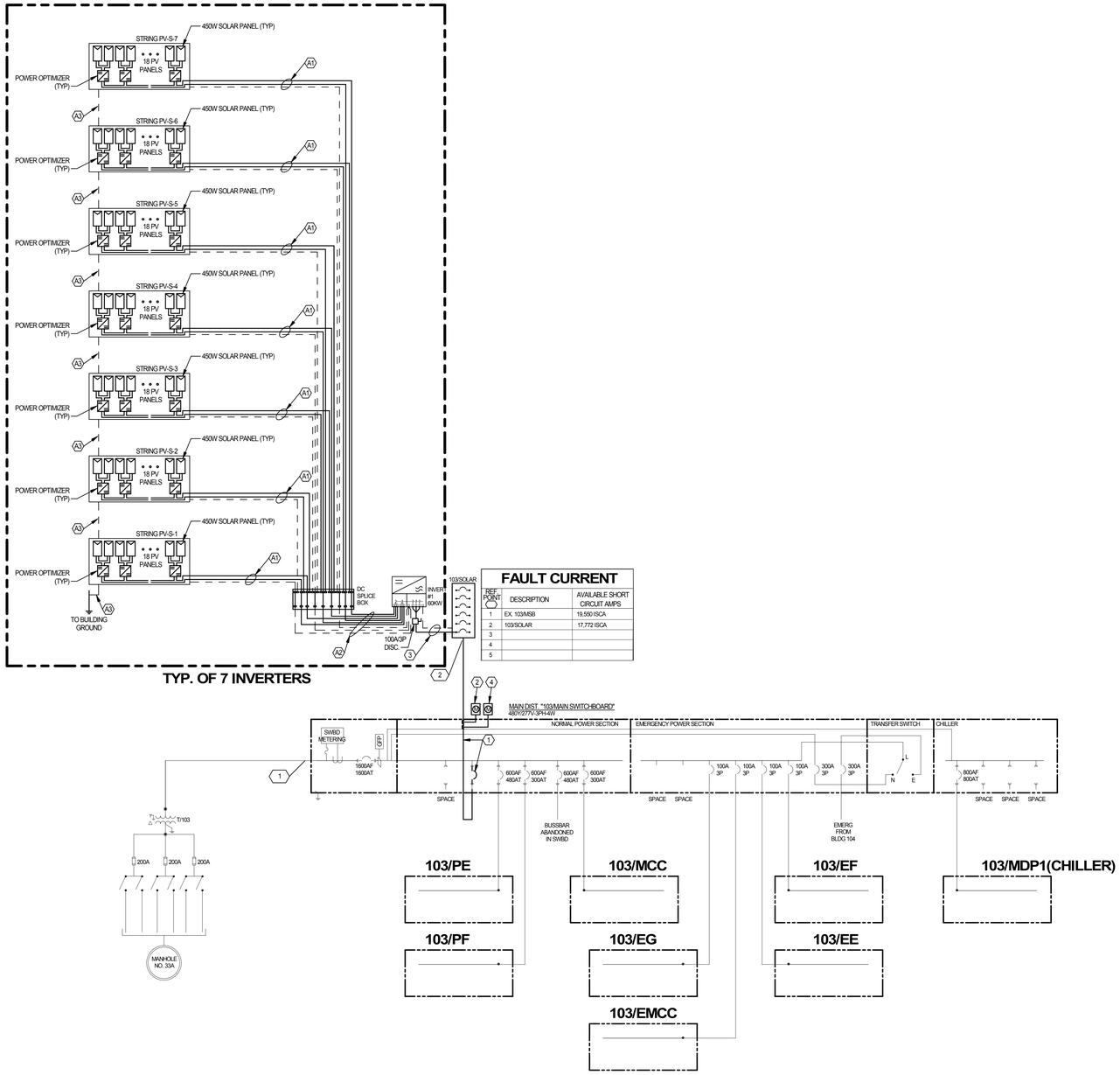
STAMP
REGISTERED PROFESSIONAL ENGINEER
MATTHEW S. GRAHAM
062-073523
STATE OF ILLINOIS
10/6/2023

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
BUILDING 104 PLAN	BID DOCUMENTS	INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	550-23-104
			Building Number: B103 & B104
Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Drawing Number: E104
		Issue Date: 10/06/2023 · Checked: JAS · Drawn: NES	

VA FORM 08 - 6231

Autodesk Docs://19.5002.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt
11/15/2023 5:07:44 PM



FAULT CURRENT

REF POINT	DESCRIPTION	AVAILABLE SHORT CIRCUIT AMPS
1	EX. 103/MSB	19,550 ISCA
2	103/SOLAR	17,772 ISCA
3		
4		
5		

1 **SINGLE LINE DIAGRAM**
SCALE: NONE

NOTES

1 NEW 600A/3P BREAKER IN EXISTING MAIN SWITCHBOARD "103/SWBD". MATCH EXISTING BREAKERS MANUFACTURER AND TYPE. RUN 2 SETS 4-350KCMIL, 1-#1 GRD. IN 4" C.

2 PROVIDE METERING TO BE MONITORED BY BAS AND CLIENT eDART SYSTEM. PROVIDE ALL REQUIRED ACCESSORIES AND PROGRAMMING. ALL WORK REQUIRING SHUT-DOWN OF ELECTRICAL SERVICES ARE TO BE PERFORMED ON A WEEKEND DAY COORDINATED WITH COR. METERING SHALL BE SIMILAR TO SCHNIEDER ELECTRIC 3500 SERIES WITH RS-485/MODBUS COMMUNICATIONS. COORDINATE INTERGRATION WITH SITE BAS MAINTENANCE VENDOR (CONSERV).

3 4-#3, 1-#8 GRD. IN 1.25" C.

4 UTILITY METERING. COORDINATE ROUGH-IN REQUIREMENTS WITH UTILITY PROVIDER.

A1 DC STRING CONDUCTOR SIZING PER MANUFACTURER RECOMMENDATIONS.

A2 NUMBER OF CONSOLIDATED DC STRING RETURNS TO INVERTER PER MANUFACTURER RECOMMENDATIONS.

A3 GEC PER MANUFACTURER RECOMMENDATIONS.

Panel: 103/SOLAR

Location: MAIN ELEC. ROOM
Supply From: 103/MSBD
Voltage: 480/277 Wye-3PH-4W
Mounting: Surface
Enclosure: Type 1
A.I.C. Rating: 22,000
Mains Type: M.L.O.
Mains Rating: 600 A

CKT	Circuit Description	Trip	Poles	A		B		C		Poles	Trip	Circuit Description	CKT
1	103/INV1	90 A	3	18.9	18.9					3	90 A	103/INV2	2
3	--	--	--			18.9	18.9			--	--	--	4
5	--	--	--					18.9	18.9	--	--	--	6
7	103/INV3	90 A	3	18.9	18.9					3	90 A	103/INV4	8
9	--	--	--			18.9	18.9			--	--	--	10
11	--	--	--					18.9	18.9	--	--	--	12
13	103/INV5	90 A	3	18.9	18.9					3	90 A	103/INV6	14
15	--	--	--			18.9	18.9			--	--	--	16
17	--	--	--					18.9	18.9	--	--	--	18
19	103/INV7	90 A	3	18.9	0.0					3	90 A	Spare	20
21	--	--	--			18.9	0.0			--	--	--	22
23	--	--	--					18.9	0.0	--	--	--	24
25	Space	--	1	--	--					1	--	Space	26
27	Space	--	1			--	--			1	--	Space	28
29	Space	--	1					--	--	1	--	Space	30
31	Space	--	1	--	--					1	--	Space	32
33	Space	--	1			--	--			1	--	Space	34
35	Space	--	1					--	--	1	--	Space	36
37	Space	--	1	--	--					1	--	Space	38
39	Space	--	1			--	--			1	--	Space	40
41	Space	--	1					--	--	1	--	Space	42
			Total Load:	132.30 kVA		132.30 kVA		132.30 kVA					

Load Classification	Connected...	Demand Factor	Estimated...	Panel Totals	
Incoming Power	396900 VA	100.00%	396900 VA		
				Total Conn. Load:	396.9 kVA
				Total Est. Demand:	396.9 kVA
				Total Conn.:	477 A
				Total Est. Demand:	477 A

Notes:

TOTAL CONNECTED	ESTIMATED DEMAND
396.9 kVA	396.9 kVA (477 A)

DESIGN NOTES:
A SYSTEM DESIGNED AROUND MULTIPLE 50KW INVERTER UNITS, EACH SERVED BY MULTIPLE STRINGS OF SOLAR PANELS. FINAL LAYOUT, NUMBER OF STRINGS, AND PANELS PER STRING TO BE COORDINATED WITH THE SOLAR EQUIPMENT SUPPLIER.

Revisions: Date:

CONSULTANT
Consultant: HEAPY
P: 937-224-0861 www.heapy.com
PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD
A/E:
GUIDON
1221 N. PENNSYLVANIA STREET
INDIANAPOLIS, IN 46202
317.800.6388

STAMP
REGISTERED PROFESSIONAL ENGINEER MATTHEW S. GRAHAM 062-073523 STATE OF ILLINOIS

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title: BUILDING 103 SINGLE LINE
Approved:
Phase: BID DOCUMENTS
FULLY SPRINKLERED
Project Title: INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS
Location: 1900 E. MAIN ST., DANVILLE, IL
Issue Date: 10/06/2023
Checked: JAS
Drawn: NES
Project Number: 550-23-104
Building Number: B103 & B104
Drawing Number: E203

VA FORM 08 - 6231

Autodesk Docs://19.50/02.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt
11/15/2023 5:07:45 PM

Panel: 104/SOLAR

Location: MAIN ELEC. ROOM
Supply From: 104/MSB
Voltage: 480/277 Wye-3PH-4W
Mounting: Surface
Enclosure: Type 1
A.I.C. Rating: 42,000
Mains Type: M.L.O.
Mains Rating: 600 A

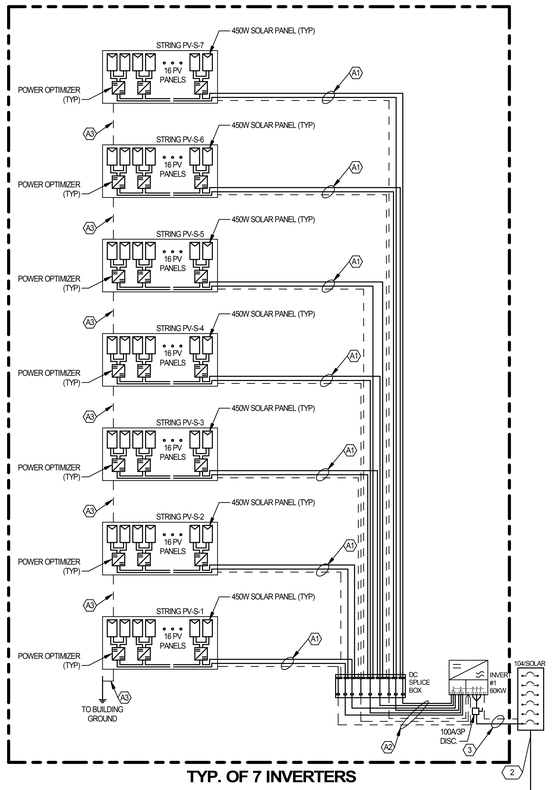
CKT	Circuit Description	Trip	Poles	A		B		C		Poles	Trip	Circuit Description	CKT
1	104/INV1	80 A	3	18900	18900					3	80 A	104/INV2	2
3	--	--	--			18900	18900			--	--	--	4
5	--	--	--					18900	18900	--	--	--	6
7	104/INV3	80 A	3	18900	18900					3	80 A	104/INV4	8
9	--	--	--			18900	18900			--	--	--	10
11	--	--	--					18900	18900	--	--	--	12
13	104/INV5	80 A	3	18900	18900					3	80 A	104/INV6	14
15	--	--	--			16.8	16.8			--	--	--	16
17	--	--	--					16.8	16.8	--	--	--	18
19	104/INV7	80 A	3	18900	18900					3	80 A	Spare	20
21	--	--	--			16.8	16.8			--	--	--	22
23	--	--	--					16.8	16.8	--	--	--	24
25	Space	--	1	--	--					1	--	Space	26
27	Space	--	1			--	--			1	--	Space	28
29	Space	--	1					--	--	1	--	Space	30
31	Space	--	1	--	--					1	--	Space	32
33	Space	--	1			--	--			1	--	Space	34
35	Space	--	1					--	--	1	--	Space	36
37	Space	--	1	--	--					1	--	Space	38
39	Space	--	1			--	--			1	--	Space	40
41	Space	--	1					--	--	1	--	Space	42
			Total Load:	132.30kVA		132.30kVA		132.30kVA					

Load Classification	Connected...	Demand Factor	Estimated...	Panel Totals	
Incoming Power	396900VA	100.00%	:396900VA		
Incoming Power	201600 VA	100.00%	201600 VA	**Total Conn. Load:**	396.9kVA
				Total Est. Demand:	396.9kVA
				Total Conn.:	477.6A
				Total Est. Demand:	477.6A

Notes:

TOTAL CONNECTED	ESTIMATED DEMAND
396.9kVA	396.9kVA (477.6A)

DESIGN NOTES:
A SYSTEM DESIGNED AROUND MULTIPLE 50KW INVERTER UNITS, EACH SERVED BY MULTIPLE STRINGS OF SOLAR PANELS. FINAL LAYOUT, NUMBER OF STRINGS, AND PANELS PER STRING TO BE COORDINATED WITH THE SOLAR EQUIPMENT SUPPLIER.



TYP. OF 7 INVERTERS

NOTES

1 UTIILZE EXISTING SPARE 800A FRAME BREAKER IN EXISTING SWITCHBOARD "104/MSB". RUN 2 SETS OF 4-350 KCMIL, 1-#1 GRD. IN 4" C.

2 PROVIDE METERING TO BE MONITORED BY BAS AND CLIENT eDART SYSTEM. PROVIDE ALL REQUIRED ACCESSORIES AND PROGRAMMING. ALL WORK REQUIRING SHUT-DOWN OF ELECTRICAL SERVICES ARE TO BE PERFORMED ON A WEEKEND DAY COORDINATED WITH COR. METERING SHALL BE SIMILAR TO SCHIENDER ELECTRIC 3500 SERIES WITH RS-485/MODBUS COMMUNICATIONS. COORDINATE INTEGRATION INTO BAS SYSTEM WITH SITE BAS SYSTEM MAINTENANCE VENDOR (CONSERV).

3 4-#3, 1-#8 GRD. IN 1.25" C.

4 UTILITY METERING. COORDINATE ROUGH-IN REQUIREMENTS WITH UTILITY PROVIDER.

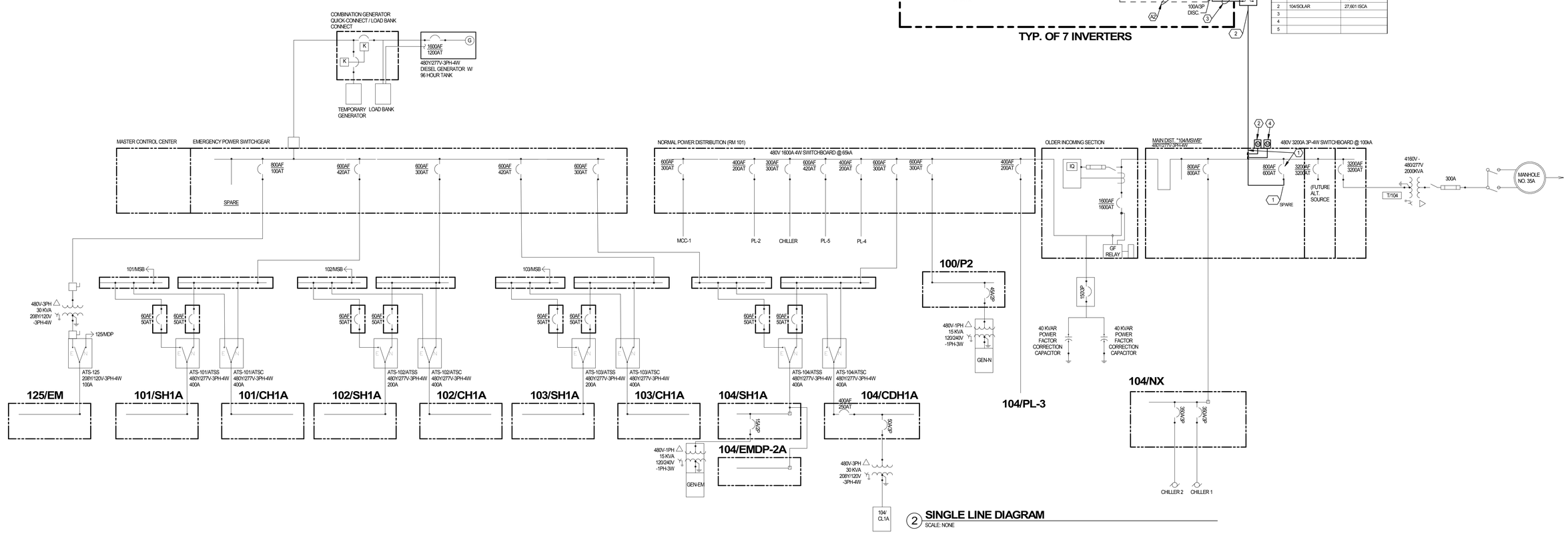
A1 DC STRING CONDUCTOR SIZING PER MANUFACTURER RECOMMENDATIONS.

A2 NUMBER OF CONSOLIDATED DC STRING RETURNS TO INVERTER PER MANUFACTURER RECOMMENDATIONS.

A3 GEC PER MANUFACTURER RECOMMENDATIONS.

FAULT CURRENT

REF. POINT	DESCRIPTION	AVAILABLE SHORT CIRCUIT AMPS
1	EX. 104/MSB	29,460 ISCA
2	104/SOLAR	27,601 ISCA
3		
4		
5		



2 **SINGLE LINE DIAGRAM**
SCALE: NONE

CONSULTANT	ARCHITECT/ENGINEER OF RECORD	STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number
Consultant: HEAPY P: 937-224-0861 www.heapy.com PROJECT NO. 2023-05127	A/E: GUIDON 1221 N. PENNSYLVANIA STREET INDIANAPOLIS, IN 46202 317.800.6388	MATTHEW S. GRAHAM 062-073523 STATE OF ILLINOIS	VA U.S. Department of Veterans Affairs	BUILDING 104 SINGLE LINE	BID DOCUMENTS	INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	550-23-104
Revisions: Date:				Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Building Number: B103 & B104
						Issue Date: 10/06/2023 · Checked: JAS · Drawn: NES	Drawing Number: E204

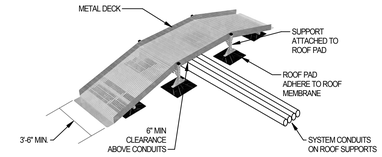
VA FORM 08 - 6231

Autodesk Docs://19.5002.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt
11/15/2023 5:07:47 PM

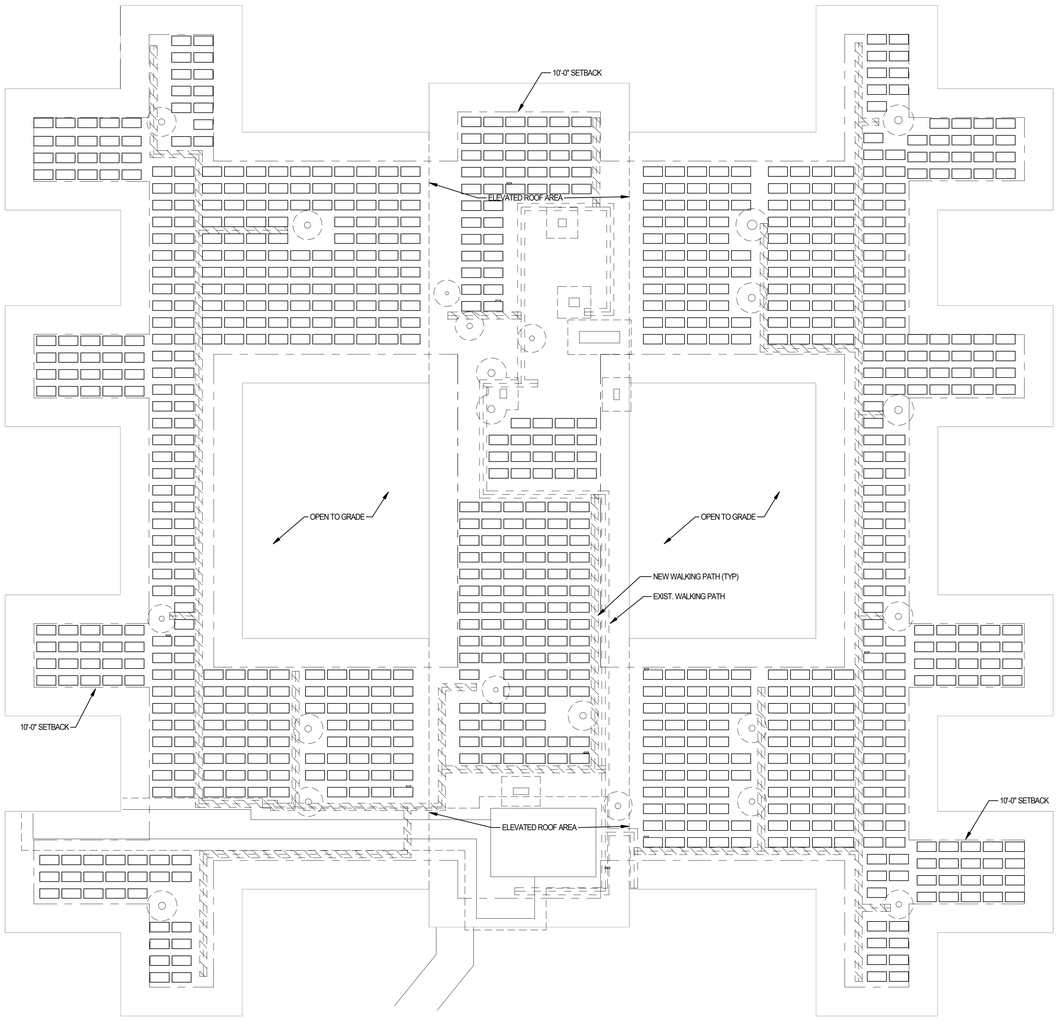
GENERAL NOTES:

A. MAINTAIN EXISTING WALKING PATHS AND PROVIDE NEW WALKWAYS AS NECESSARY ALONG ROOF FOR ACCESS TO EXISTING ROOF MOUNTED EQUIPMENT. MAINTAIN A WALKING PATH CLEARANCE WIDTH MINIMUM OF 3'-6".

B. ROUTING OF THE SOLAR ARRAY SYSTEMS CONDUITS SHALL NOT CROSS OR IMPED ACCESS TO WALKING PATHS. WHERE THIS IS NOT POSSIBLE, PROVIDE CROSS OVER OR RAMP SYSTEM TO MAINTAIN A CLEAR WALKING PATH ON ROOF. CROSS OVER/RAMP SHALL BE SIMILAR TO MIRO INDUSTRIES BRIDGE CROSSOVER.

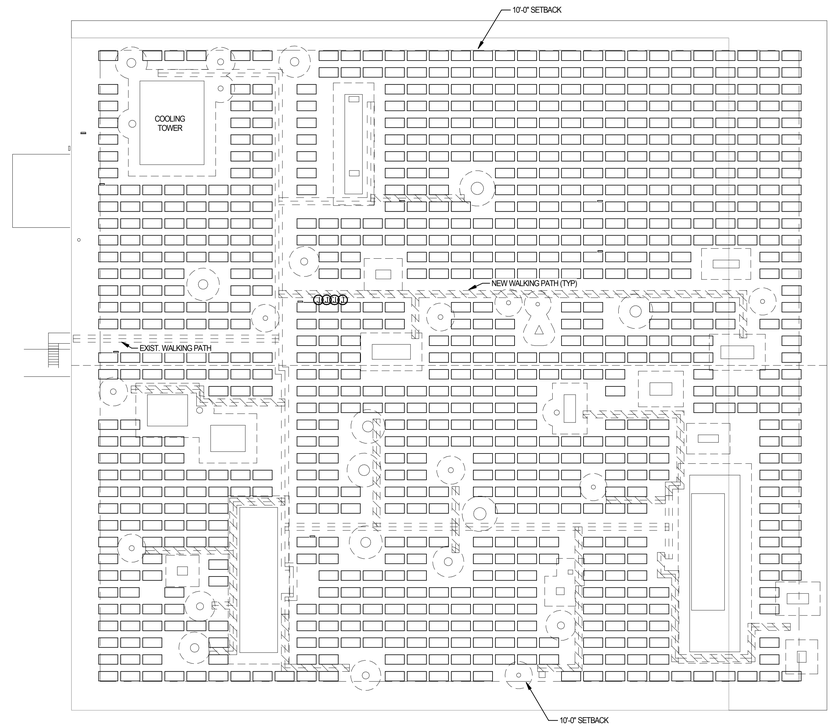


3 ROOF MOUNTED BRIDGE



1 BUILDING 103 WALKING PATH PLAN

SCALE: 3/64" = 1'-0"



2 BUILDING 104 WALKING PATH PLAN

SCALE: 3/64" = 1'-0"

Revisions:	Date:

CONSULTANT

Consultant:

HEAPY

P: 937-224-0861 www.heapy.com

PROJECT NO. 2023-05127

ARCHITECT/ENGINEER OF RECORD

A/E:

GUIDON

1221 N. PENNSYLVANIA STREET

INDIANAPOLIS, IN 46202

317.800.6388

STAMP

REGISTERED PROFESSIONAL ENGINEER MATTHEW S. GRAHAM 062-073523 STATE OF ILLINOIS

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title	Phase	Project Title	Project Number
ROOF PATHING PLANS	BID DOCUMENTS	INSTALL ROOFTOP PHOTOVOLTAIC - MULTIPLE BUILDINGS	550-23-104
			Building Number: B103 & B104
Approved:	FULLY SPRINKLERED	Location: 1900 E. MAIN ST., DANVILLE, IL	Drawing Number: E301
		Issue Date: 10/06/2023 · Checked: · Drawn:	

VA FORM 08 - 6231

Autodesk Docs://19.5002.17 - Install Rooftop Photovoltaic/2023-05127-Electrical-Central.rvt

11/15/2023 5:07:57 PM